

RESOURCE

Dynamic and rapid changes in the transcriptome and epigenome during germination and in developing rice (*Oryza sativa*) coleoptiles under anoxia and re-oxygenation

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SUMMARY

Detailed molecular profiling of *Oryza sativa* (rice) was carried out to uncover the features that are essential for germination and early seedling growth under anoxic conditions. Temporal analysis of the transcriptome and methylome from germination to young seedlings under aerobic and anaerobic conditions revealed 82% similarity in the transcriptome and no differences in the epigenome up to 24 h. Following germination, significant changes in the transcriptome and DNA methylation were observed between 4-day aerobically and anaerobically grown coleoptiles. A link between the epigenomic state and cell division versus cell elongation is suggested, as no differences in DNA methylation were observed between 24-h aerobically and anaerobically germinating embryos, when there is little cell division. After that, epigenetic changes appear to correlate with differences between cell elongation (anaerobic conditions) versus cell division (aerobic conditions) in the coleoptiles. Re-oxygenation of 3-day anaerobically grown seedlings resulted in rapid transcriptomic changes in DNA methylation in these coleoptiles. Unlike the transcriptome, changes in DNA methylation upon re-oxygenation did not reflect those seen in aerobic coleoptiles, but instead, reverted to a pattern similar to dry seeds. Reversion to the 'dry seed' state of DNA methylation upon re-oxygenation may act to 'reset the clock' for the rapid molecular changes and cell division that result upon re-oxygenation.

Keywords: low oxygen, rice, RNA-seq, DNA methylation.

INTRODUCTION

Seed germination represents a critical stage in a plant's life cycle. It represents a large developmental transition from dormancy to an active metabolic state in a relatively small time period, in which the storage reserves in the seed must be utilised for seeding establishment. Germination and early seedling development in plants represent stages with very high energy demands, with plants generally relying on oxygen to produce the ATP necessary for this developmental progression (Edwards *et al.*, 2012). Oxygen deprivation, often caused by flooding, can ultimately result in significant crop loss (Voesenek *et al.*, 2006). Different plant species have a range of tolerances to these conditions, with species of rice (*Oryza sativa*) having the ability to

germinate and develop in the complete absence of oxygen. It is important to note that while rice is known to be resistant to flooding, the underlying molecular features that dictate tolerance to anoxia differ between different developmental states. Rice readily germinates under anaerobic conditions (Mageschi and Perata, 2008), with a trehalose-6-phosphate phosphatase identified to enhance anaerobic germination in rice (Kretschmar *et al.*, 2015), possibly by acting as an energy sensor enhancing starch mobilisation to drive growth. During vegetative growth rice utilises other strategies to maintain tolerance to submergence. Rapid growth and elongation of the anaerobically grown coleoptile is referred to as the 'snorkel effect'

(Kordan, 1974), and molecular analyses have revealed key players (the *SNORKEL1* and -2 genes) that have a role in this mode of submergence tolerance (Hattori *et al.*, 2009). An alternative quiescence strategy is observed in rice varieties with the *SUB1A-1* locus. Lines with *SUB1A-1* limit the mobilisation of starch and thus produce less ethanol and other products from fermentation as well as affecting senescence-related processes (Fukao *et al.*, 2012). Escape and quiescence strategies have been observed to be utilised by plant species under flooding conditions (Voisenek *et al.*, 2006).

In the last decade, the involvement of group VII ethylene responsive factors in tolerance of low oxygen has also been revealed across different species (Voisenek and Sasidharan, 2013). As outlined above in rice, the identification of *SUB1A* (Xu *et al.*, 2006) and the *SNORKEL 1/2* genes (Hattori *et al.*, 2009) revealed key factors involved in tolerance of submergence. Similarly, the more recent identification of the ethylene response factors RAP2.12/RAP2.2 and HRE1/2 in Arabidopsis confirmed an important role for these factors in successfully improving tolerance of low oxygen (Hinz *et al.*, 2010; Licausi *et al.*, 2010). A low-oxygen sensing system based on the N-end rule has also been identified in Arabidopsis, where under normoxic conditions a cysteine residue undergoes oxidation and is targeted for breakdown through the proteasome system, but under anoxic conditions it is stabilised and relocated to the nucleus via an unknown mechanism (Gibbs *et al.*, 2011). In addition, studies are also revealing important roles for signalling by nitric oxide (NO) and reactive oxygen species (ROS) via the mitochondrial respiratory components under low-oxygen conditions (Stoimenova *et al.*, 2007; Blokhina and Fagerstedt, 2010; Igamberdiev *et al.*, 2010). The identification of Hypoxia Responsive Universal Stress Protein 1 (HRU1) as a key mediator of ROS homeostasis under anoxic conditions reveals that a ROS signalling network is required for tolerance of low oxygen in Arabidopsis (Gonzali *et al.*, 2015).

A variety of biochemical and molecular changes have been characterised in plants subjected to flooding conditions. One of the earliest and most common molecular responses to low oxygen is the induction of fermentation and glycolysis pathways, which is seen across both low-oxygen-tolerant and -intolerant plants. In rice, this involves increased fermentation, as well as modified starch and sucrose metabolism, feeding the glycolytic pathway. Using mutants for alcohol dehydrogenase 1 (*ADH1*) in rice, it has been shown that ADH1 is required for cell division and elongation in the anoxic rice coleoptile (Takahashi *et al.*, 2011). Additionally, a switch from ATP to pyrophosphate (PPI)-linked glycolytic enzymes has also been observed under anoxia in rice (Huang *et al.*, 2008; Magneschi and Perata, 2008), with meta-analysis suggesting that specific isoforms of ATP-dependent phosphofructokinase (PFK)

and the pyrophosphate (PPI)-dependent pyrophosphate-fructose-6-phosphate phosphotransferase (PFP) contribute to the altered glycolytic activity observed under anoxia (Mustroph *et al.*, 2013). While these genes may contribute to tolerance of low oxygen in rice, altering fermentation enzymes and glycolytic enzymes alone does not confer tolerance in anoxia-intolerant species (Banti *et al.*, 2013).

In addition to the transcript, protein and metabolite levels, recent studies have also revealed molecular changes at the DNA level, in particular changes in DNA methylation in response to stress (Secco *et al.*, 2015) as well as during germination (Yamauchi *et al.*, 2014) and seed development (Xing *et al.*, 2015). In plants, DNA methylation occurs in the context of three sequences, namely CG, CHG and CHH (where H = A, C or T), which are maintained through specific DNA methyltransferases (Matzke and Mosher, 2014). Specifically, DNA methyltransferase 1 (MET1) and chromomethylase 3 (CMT3) are known to be responsible for maintaining DNA methylation in the CG and CHG contexts, respectively, while *de novo* methylation and methylation in the CHH context is carried out by domains rearranged methyltransferases (DRM1/2) and chromomethylase 2 (CMT2) (Du *et al.*, 2012; Zemach *et al.*, 2013; Matzke and Mosher, 2014). The loss of function for a number of these genes (e.g. *OsDRM2*, *OsCMT3*, *OsDCL3* and *OsMET1*) in rice has recently been shown to result in severe developmental phenotypes, particularly affecting fertility (Moritoh *et al.*, 2012; Hu *et al.*, 2014; Wei *et al.*, 2014; Cheng *et al.*, 2015). For example, mutation in a major CG maintenance DNA methyltransferase (*OsMet1-2*; LOC_Os07 g08500) has been shown to have abnormal seed phenotypes and display embryonic and seedling lethality (Hu *et al.*, 2014; Yamauchi *et al.*, 2014), while *OsDRM2* (LOC_Os03 g02010) mutants have been shown to display abnormal developmental defects including sterility (Moritoh *et al.*, 2012). While such studies examining the loss of function of individual genes have offered much insight into each context of methylation (CG, CHG and CHH), and have clearly shown an essential role for changes in DNA methylation during germination and development, the genome-wide changes at these stages have not been investigated.

The agronomic value of anaerobic tolerance should not be underestimated. The breeding of the *SUB1* of rice into elite agricultural varieties in flood-prone regions in Asia has made a major contribution to food security (Voisenek and Bailey-Serres, 2015). Increased anaerobic tolerance in rice is required to maintain and increase rice yields (Kretschmar *et al.*, 2015). Thus, an understanding at a molecular level of all the changes occurring in both the transcriptome and in DNA methylation is needed to reveal the molecular basis of why alteration of DNA methylation has such a dramatic impact on seedling viability and development in rice (Moritoh *et al.*, 2012; Hu *et al.*, 2014; Wei *et al.*, 2014; Yamauchi *et al.*, 2014; Cheng *et al.*, 2015),

and if it differs between aerobic and anaerobic conditions. We hypothesised that epigenomic changes have an important role during germination and sought to comprehensively examine changes in DNA methylation and the transcriptome during seed germination and early development under both aerobic and anaerobic conditions as well as upon re-oxygenation. These findings not only provide insight into the epigenetic changes seen during this crucial developmental stage, but also reveal unique patterns of rapid oxygen-dependent variation in DNA methylation, suggesting a role for epigenetic modification in these important processes.

RESULTS

Rice development and experimental design

To gain insight into the molecular responses to oxygen deprivation, an extensive time course was examined, from dry seeds to germinating embryos (24 h) and up to 4-day-old rice coleoptiles, as well as coleoptiles collected from seedlings grown for 3 days under anoxic conditions and transferred to air for 1 day (Figure 1a). Seeds were sterilised and then transferred into liquid media infused with a constant flow of air (A) or nitrogen gas (N) in the dark. Total RNA was isolated from dry seed embryos and germinating embryos up to 24 h, after which only coleoptile tissue was collected. RNA sequencing (RNA-seq) libraries were generated from rRNA-depleted total RNA using three independent replicates per time point and a total of 48 libraries were sequenced. For the dry seed embryos, 24-h embryos, 4-day coleoptiles and 3dN1dA samples, DNA was also extracted and 18 libraries were generated for whole-genome bisulphite sequencing (WGBS) by MethylC-seq (Urich *et al.*, 2015) (Figure 1a). As expected, rice development differed under aerobic and anaerobic conditions, with more of the morphological differences becoming apparent after 24 h (Figure 1b). Under anoxia, coleoptile elongation occurs without radicle emergence; however, within only 1 day of re-oxygenation, further coleoptile elongation and significant radicle emergence is observed in the seedling (3dN1dA; Figure 1b).

Transcriptome responses to anoxia

In order to identify any unannotated transcripts or gene isoforms in the RNA-seq data over the time course, a reference annotation-based transcript (RABT) assembly was generated using the Michigan State University (MSU) Rice Genome Annotation Project assembly (TIGRv7) as reference. In this way, thousands of significantly differentially expressed genes (DEGs) were identified [$P < 0.05$, q -value < 0.05 = false discovery rate (FDR) $< 5\%$] during germination (A versus N), early seedling development (A versus N) and after re-oxygenation (4dN versus 3dN1dA) (Figure 2a). Interestingly, the number of DEGs in coleoptile tissues was

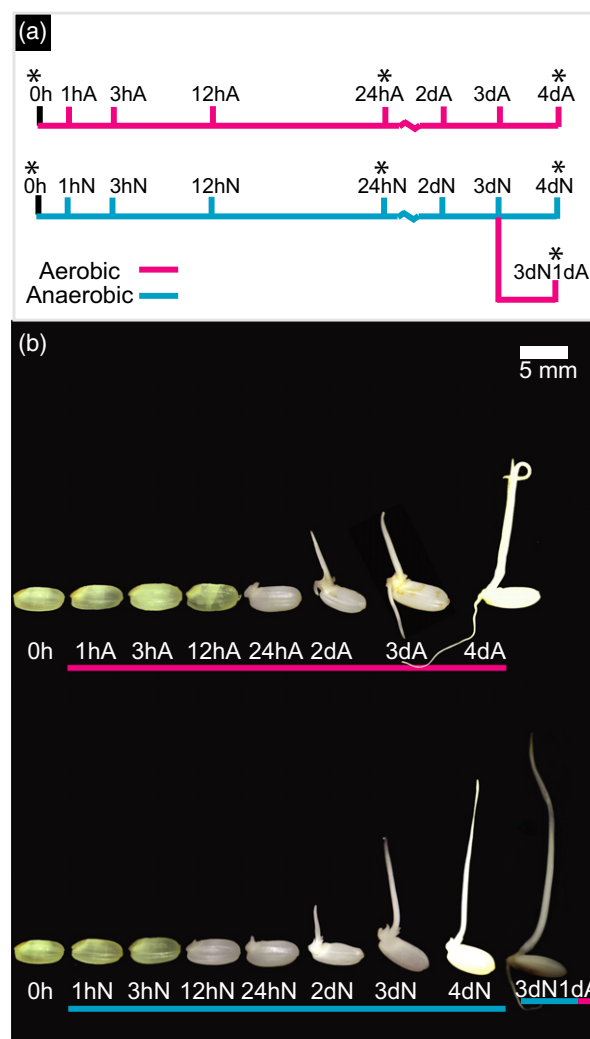


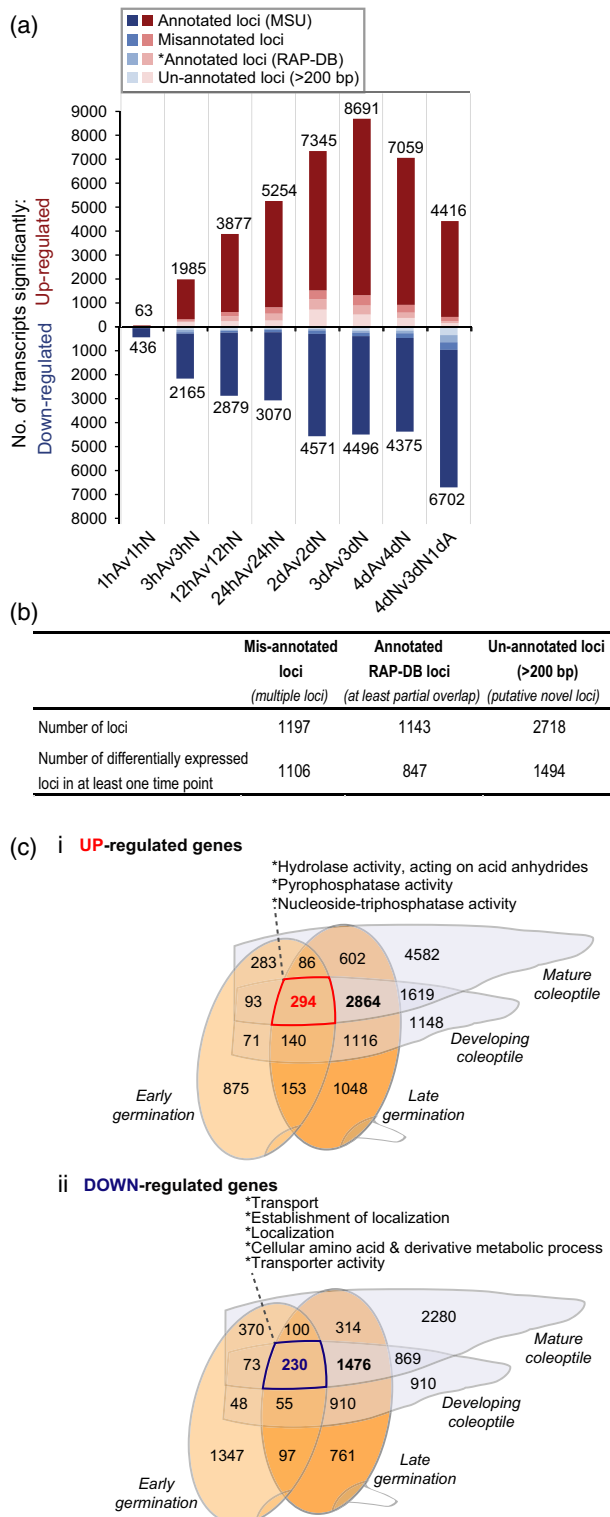
Figure 1. Rice development and experimental design.

(a) Schematic representation of the experimental design. Seeds were sterilised and then transferred into liquid media infused with a constant flow of air (A) or nitrogen gas (N). Embryos were collected from dry seeds and up to 2 days (2 d), after which only the primary coleoptiles were collected (3-day and 4-day samples). Asterisks (*) indicate samples that were also analysed by DNA methylome sequencing.

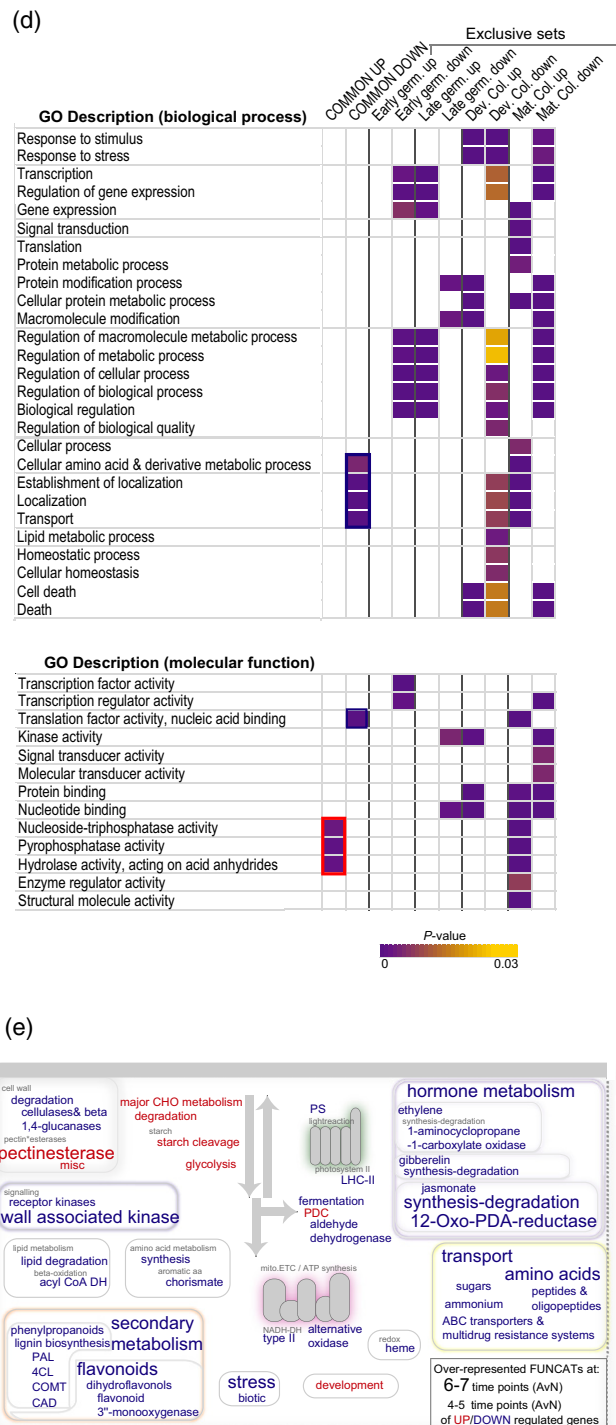
(b) Rice development under aerobic (A) and anaerobic (N) conditions from dry seeds to 4-day-old (4 d) seedlings. An example of a seed/seedling from each of the time points are shown, including one grown under anaerobic conditions for 3 days and transferred to air for 1 day (3dN1dA).

60% larger than the number of DEGs at one or more time points during germination (A versus N; Figure 2a), which also corresponded with the more distinct morphological differences observed following germination under differing oxygen conditions (Figure 1b).

The RNA-seq data for each sample can be visualised using the AnnoJ genome browser at http://www.plantenergy.uwa.edu.au/annoj/Reena_Rice_AN_RNA_mC.html. For each time point, corresponding numbers of TIGR7 annotated, mis-annotated and unannotated loci were identified



(Figure 2a,b and Figure S1), where mis-annotated loci refers to those identified using the RABT assembly but spanning two or more loci in the TIGRv7 annotation (Figure S1a). The chromosomal locations of the unannotated loci identified by the RABT assembly were first



overlapped with the Rice Annotation Project Database (RAP-DB) annotation, revealing that some of these were known genes (*Annotated loci – RAP-DB; Figures 2a,b and S1a). In this way, the 1494 differentially expressed, unannotated loci identified here represent regions that do

Figure 2. Transcriptomic responses to oxygen deprivation.

(a) RNA sequencing analysis of differential expression using a reference annotation-based transcript assembly. Corresponding numbers of TIGR7 (MSU) annotated, mis-annotated and unannotated loci are shown. *Note that all unannotated loci were overlapped with Rice Annotation Project Database (RAP-DB) annotated loci and the overlapping numbers are shown.

(b) Summary of the numbers of mis-annotated, overlapping RAP-DB annotated and unannotated loci.

(c) The anaerobic responses were compared at four developmental stages as follows: early germination [1hA versus 1hN and/or 3hA versus 3hA; A, aerobic (air) conditions; N, anaerobic (nitrogen) conditions], late germination (12hA versus 12hN and/or 24hA versus 24hN), developing coleoptile (2dA versus 2dN; d, days) and mature coleoptile (3dA versus 3dN and/or 4dA versus 4dN). The numbers of genes showing overlapping responses are indicated within the Venn diagrams for (i) upregulated genes and (ii) downregulated genes.

(d) Gene Ontology (GO) over-representation analysis was carried out on common upregulated genes and common downregulated genes as well as each of the exclusive gene sets, e.g. the 761 exclusive genes that were downregulated during late germination (Late germ. down) were significantly enriched ($P < 0.03$) in protein modification and macromolecule modification functions. P -values are shown as a heatmap with the lowest P -values being shown in purple. (E) Over-representation analysis was carried out comparing aerobic with anaerobic transcriptomes (A versus N) at each of the seven time points analysed, from germination (1, 3, 12 and 24 h) to the developing and mature coleoptile (2, 3 and 4 days). Significantly over-represented functional categories (Fisher's test, $P < 0.05$) that were identified in at least four time points are shown for all functions involved in metabolism across the sets of upregulated genes (red font) and downregulated genes (blue font). Font size reflects the level of conservation for the over-representation of that functional category, e.g. genes encoding stress-related functions were significantly over-represented in the downregulated genes across the entire time course (seven time points).

not overlap with known TIGR7-MSU and/or RAP-DB annotated loci (Figure 2b). Of these, 91 loci showed greater than fourfold differential expression in one or more (A versus N) comparisons at each time point, and an MSU-BLAST analysis revealed that 78 loci showed significant similarity ($E\text{-value} < 1 \times 10^{-10}$) with regions of known annotated genes (Figure S2a). For example, the putative unannotated gene XLOC_045312 was induced during oxygen deprivation and showed significant similarity to a known MYB transcription factor, which was also induced (LOC_Os04 g51800; Figure S2b–d). Similarly, XLOC_014344 was downregulated under anoxia and showed significant similarity to a known downregulated nucleoside triphosphatase (LOC_Os11 g25260; Figure S3). While XLOC_014344 showed the greatest homology to LOC_Os11 g25260, it was downstream of another downregulated nucleoside triphosphatase LOC_11 g25330 (Figure S3). Given the relatively small size of XLOC_014344 (307 bp), it is also possible that XLOC_014344 may represent an unannotated exon from another nucleoside triphosphatase.

Stage-specific and conserved transcriptomic responses to oxygen deprivation

Previous microarray studies in rice have analysed the transcriptomes of 24-h embryos (Narsai *et al.*, 2009) and 4-day coleoptiles (Lasanthi-Kudahettige *et al.*, 2007) under anoxia. More than double this number were identified for the equivalent time points in this study, providing a more complete view of germination and early seedling growth in rice (Figure 2a). To identify tissue/stage-specific responses, all time points were placed into developmental stages as follows: early germination (1hA versus 1hN and/or 3hA versus 3hA), late germination (12hA versus 12hN and/or 24hA versus 24hN), developing coleoptile (2dA versus 2dN) and mature coleoptile (3dA versus 3dN and/or 4dA versus 4dN). Genes that were responsive at one or more time point at each of the stages were identified and overlapped across the different developmental stages (Figure 2c). Significant overlap was observed

(2864 upregulated genes and 1476 downregulated genes) between late germination and coleoptile tissues (Figure 2c). Of all the upregulated genes during early germination, 44% (875 genes) were responsive only at this stage, similarly 44% (4582 genes) of all the upregulated genes in the mature coleoptile were responsive only at this stage (Figure 2ci). A similar trend is observed for the downregulated genes, with an even larger percentage (1347 genes; 58%) of genes showing an exclusive response during early germination (Figure 2cii). In contrast, only 17% and 16% of the upregulated genes, and similar percentages of the downregulated genes during late germination or the developing coleoptile, respectively, were exclusive (Figure 2c). The 294 upregulated genes across all stages showed a significant enrichment (AgriGO, $P < 0.05$, FDR < 0.05) in the Gene Ontology (GO) molecular function of hydrolase activity, pyrophosphatase activity and nucleoside triphosphatase activity including several ATP-binding proteins such as RNA helicases (LOC_Os09 g34910, LOC_Os12 g41715). In contrast, the 230 common downregulated genes were significantly enriched in transport and localisation functions, including several potassium and sulphate transporters (e.g. LOC_Os03 g37840, LOC_Os08 g31410) (Figure 2cii).

The functions of the genes showing exclusive responses at each stage were examined to identify possible tissue-specific changes in gene expression. The largest set of genes showing a tissue/stage-specific response is for the mature coleoptile, with signal transduction, translation and protein metabolism over-represented in the exclusive set of upregulated genes (Figure 2d). Among these, 190 genes (out of the 232 DEGs) encoding ribosomal proteins were induced in the mature anoxic coleoptile and not at any other stage over the time course, indicating that the upregulation of translation and protein synthesis is specific to the coleoptile grown under anoxic conditions. It is worth noting that despite being identified as exclusive, there were also some similarities in the over-represented functional categories of these exclusive gene sets (Figure 2d). For example, transcription, regulation of gene expression

and other regulatory functions were over-represented in the downregulated gene set during early germination as well as in the developing and mature coleoptile (Figure 2d). In contrast, these functions were enriched in the upregulated genes during late germination (Figure 2d). Investigation of sub-groups in these sets, such as individual transcription factor (TF) families, may unveil differences in the expression of particular families over anaerobic development. Similarly, while the regulation of metabolism is over-represented at multiple stages during anaerobic development (Figure 2d), further investigation of sub-categories may reveal differences in the expression of genes encoding specific aspects of metabolism.

Pageman functional over-representation analysis (Usadel *et al.*, 2006) revealed genes encoding specific metabolic functions that were consistently induced or repressed over germination and in the coleoptile under anoxia. Not surprisingly, the induced genes were significantly enriched in carbohydrate metabolism, particularly starch degradation (cleavage), glycolysis and pyruvate decarboxylase (fermentation) (Figure 2e). Two genes encoding proteins involved in starch degradation (starch-binding D enzyme, LOC_Os05 g37450) and glycolysis (phosphoglucosmutase, LOC_Os10 g11140) were among the 294 common upregulated genes under anaerobic conditions (Figure 2ci). These findings not only support previous studies (Narsai *et al.*, 2009), but also reveal other genes/pathways with altered expression under these conditions (Figure 2e).

Notably, many more functional categories showed conserved patterns of repression rather than induction under anaerobic conditions (Figure 2e). Genes encoding stress responsive functions showed the greatest conservation among the downregulated gene sets, with an over-representation of these observed at all seven time points from 1-h to 4-day coleoptiles (Figure 2e). Similarly, genes encoding transport functions and hormone metabolism were reduced in expression (Figure 2e). Interestingly, despite the upregulation of pyruvate decarboxylase, aldehyde dehydrogenase-encoding genes were significantly downregulated, with one of these genes (LOC_Os01 g40870) down-regulated from as early as 1 h, through to the 4-day-old coleoptile under anoxia (common downregulated gene; Figure 2cii). Similarly, the genes encoding type II NADH dehydrogenase, the alternative oxidase and non-symbiotic haemoglobin (heme) were down-regulated over anaerobic development (Figure 2e), with a gene encoding an alternative oxidase (LOC_Os04 g51150) also identified as a common downregulated gene (Figure 2cii). Overall, both common and stage-specific differences can be observed in the expression of gene sets encoding different functional pathways under oxygen deprivation, with the largest transcriptomic response observed in mature coleoptiles (Figure 2c).

Changes in DNA methylation under oxygen deprivation and upon re-oxygenation

Whole-genome single-base resolution profiling of DNA methylation by MethylC-seq for dry seeds (0 h), germinated seeds (24hA versus 24hN), mature coleoptiles (4dA versus 4dN) and following re-oxygenation (4dN versus 3dN1A) revealed differentially methylated regions (DMRs) between some of these comparisons (Figure 3). Only regions that showed significant differential methylation using all three replicates from each sample were considered for further analysis (FDR < 0.01). Interestingly, no significant DMRs were identified between the 24-h germinated seedling under anoxia compared with aerobic conditions (24hA versus 24hN; Figure 3). However, 256 and 282 DMRs were identified under oxygen deprivation in the 4-day coleoptile (4dA versus 4dN) and following re-oxygenation (3dN1dA versus 4dN; Figure 3). When these two sets were compared, they had only one DMR in common (Figure 3). This suggests that the changes in DNA methylation, which occur upon re-oxygenation are largely unique to this treatment and do not simply represent a reversal in the pattern of methylation observed under anoxia. When each of the DMRs was assigned to the nearest transcript, most (>70%) of these were within a distance of 3 kb of the 80 and 55 DEGs identified under anoxia and re-oxygenation, respectively (Figure 3).

When methylation levels were analysed in all sequence contexts (CNN) for each DMR, it was seen that all of the 256 DMRs identified in the 4-day coleoptile were hypermethylated under oxygen deprivation (4dA versus 4dN; Figure S4). Interestingly, there does appear to be an overall reduction in the methylation levels after re-oxygenation; however, this was not statistically significant (FDR < 0.01) (3dN1dA; Figure 4a). Hierarchical clustering of the percentage difference in methylation between anaerobic and aerobic conditions in each context (CG, CHG and CHH) revealed

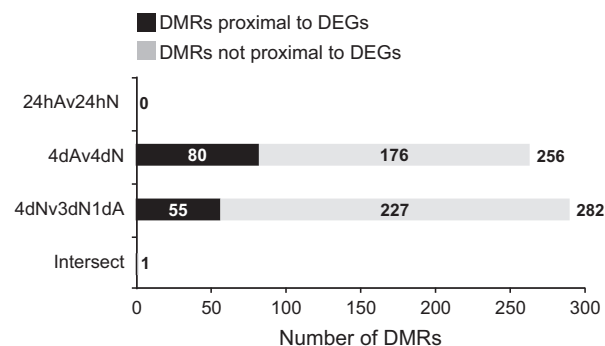


Figure 3. Summary of the number of differentially methylated regions (DMRs) in response to oxygen deprivation and re-oxygenation. For each of the DMRs, proximal genes were also identified and the number of DMRs proximal to differentially expressed genes (DEGs) are indicated in black (A, aerobic/air; N, anaerobic/nitrogen; v, versus; d, days; h, hours).

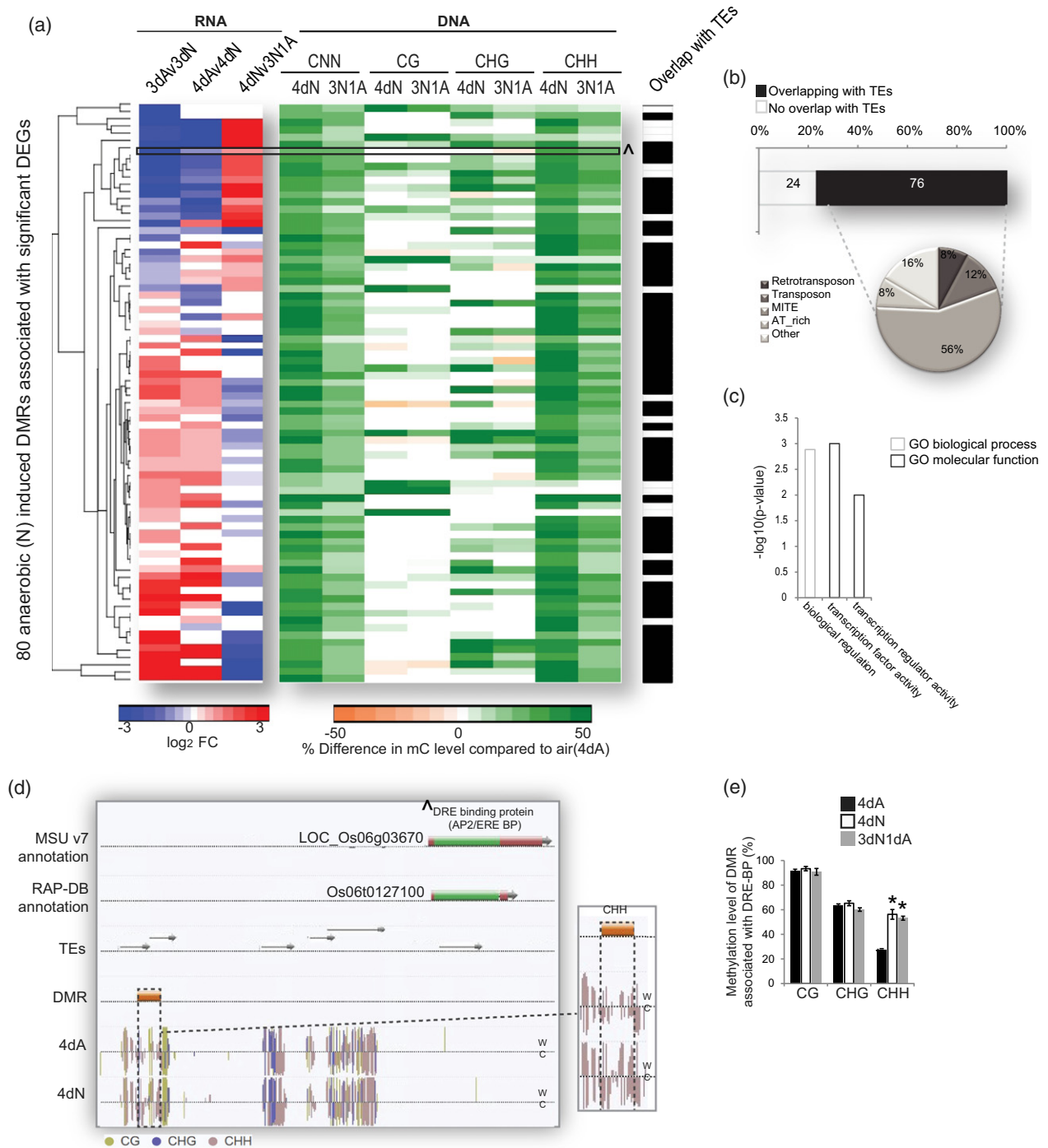


Figure 4. Differential methylation between the aerobically and anaerobically grown coleoptiles.

(a) Of the 256 differentially methylated regions (DMRs) between 4-day aerobically (4dA) versus 4-day anaerobically (4dN) grown coleoptiles, 80 were proximal to genes showing differential expression. Fold-changes (FC) of these 80 genes were hierarchically clustered and shown as a heatmap. Next to this, Cytosine DNA methylation (mC) levels are shown as percentage differences compared with 4dA as a heatmap. DMRs overlapping with transposable elements (TEs) are indicated by black boxes.

(b) The percentage of DMRs overlapping with TEs is shown, as well as the breakdown of the types of TEs.

(c) Functional categories identified by Gene Ontology (GO) over-representation analysis of the 80 proximal DEGs.

(d) An example of a differentially expressed gene (4dA versus 4dN) encoding a dehydration response element-binding protein (DRE-BP; LOC_Os06 g03670), which is proximal to a DMR in the upstream region of the gene.

(e) Methylation levels in the different contexts (CG, CHG, CHH) in the 4-day coleoptiles (4dA and 4dN) and upon re-oxygenation (3dN1dA) for the DMR associated with DRE-BP (LOC_Os06 g03670). Significant differences compared with 4dA are indicated by an asterisk (*) (where $P < 0.05$; Student's *t*-test).

that the hyper-methylation largely occurred in the CHH context, with a subset of DMRs showing changes in CG and CHG methylation (Figure S4). When these DMRs were overlapped with genomic regions containing repetitive sequences and transposable elements (TEs), 79% of the 256 anaerobically induced DMRs overlapped (Figure S4). It is also apparent that when the 80 anaerobically induced DMRs associated with DEGs were examined further these patterns were conserved (Figure 4a,b). Both the 3-day and 4-day comparisons of transcript abundance (A versus N) were included in identifying the set of 80 DEGs under anaerobic conditions and all fold-changes were hierarchically clustered, with the differences in methylation shown next to this (Figure 4a). Fifty of the 80 DEGs associated with DMRs were significantly induced ($P < 0.05$, q -value $< 0.05 = \text{FDR} < 5\%$) under anaerobic conditions, and all of the nearby DMRs were hyper-methylated compared with aerobic conditions, predominantly in the CHH context (AvN; Figure 4a). Additionally, 76% of the DMRs associated with DEGs overlapped with TEs, with 56% representing miniature inverted repeat transposable elements (MITEs), while retrotransposons and transposons made up another 20% (Figure 4b).

When the 80 DEGs proximal to DMRs were analysed to determine if these were over-represented in a specific function, regulation and TF functions were significantly enriched, with 11 of the 80 DEGs encoding TFs/regulators (Figure 4c). Some examples of these included an ethylene-responsive TF (LOC_Os04g32620), a bZIP factor (LOC_Os12g09250) and a dehydration-responsive element (DRE)-binding protein (LOC_Os06g03670) (Figure 4c). Previous studies in both *Arabidopsis* and rice have shown a link between ethylene-responsive factors and survival under anaerobic conditions (Hattori *et al.*, 2009; Voesenek and Sasidharan, 2013). When the gene encoding the ethylene-responsive DRE-binding protein (LOC_Os06g03670) was examined more closely, it was downregulated under anoxia and significantly induced after re-oxygenation (black box[^] in Figure 4a). Additionally, a significant DMR was observed upstream of this gene, overlapping with two known TE regions (Figure 4d). A significant difference in DNA methylation was observed ($\text{FDR} < 0.01$), particularly in the CHH context, with an increase of about 40% in methylation observed under anaerobic compared with aerobic conditions in the 4-day coleoptile (Figure 4d,e). Notably, this hyper-methylation was still not significantly reduced after re-oxygenation at 3dN1dA ($\text{FDR} < 0.01$) (Figure 4e).

Specific responses of TFs and DNA methyltransferases under oxygen deprivation and re-oxygenation

The over-representation of genes encoding TFs was not only seen in the 80 DEGs proximal to DMRs (Figure 4c) but also in the DEGs at other stages (Figure 2d). The Plant Transcription Factor Database (<http://plntfdb.bio.uni-potsda>

m.de/v3.0/) was used to classify the genes encoding rice TFs in this study, and an over-representation analysis of these (Fishers test, $P < 0.05$) revealed stage-specific and conserved changes in the different families of TFs (Figure 5a). The most conserved changes were the significant over-representation of TIFY and WRKY families (Fisher's test, $P < 0.05$) in the downregulated genes under anoxia and upregulated genes in response to re-oxygenation (Figure 5a). In contrast, the bZIP and HSF families were over-represented in the genes induced under anoxia (Figure 5a). Notably, of the fewer than 500 DEGs at 1 h under anoxia (Figure 2a), there was a significant enrichment (Fisher's test, $P < 0.05$) of TIFY, WRKY and AP2-EREBP TFs (Figure 5a). While most of the aforementioned TF families showed similarities in expression over the time course, other TF families showed stage-specific responses and differences within the families (Figure 5a). For example, AP2-EREBPs were over-represented in the upregulated genes at 3 h, 24 h and 2 days, as well as the downregulated genes at 1 h, 3 h and 3 days under anoxia (Figure 5a). Thus, it appears that both stage-specific and conserved regulation of several TF families occurs from as early as 1 h under anoxia (Figure 5a).

To reveal the TFs with the most conserved gene expression responses to anoxia, we identified 32 genes that were differentially expressed at seven of the eight time points (including re-oxygenation) in this study (Figure 5b). All of these were significantly responsive to 1 day of re-oxygenation in a manner opposite to that observed under anaerobic conditions (Figure 5b). Overall, there was a trend for the downregulation of TF-encoding genes during early germination (1 h and 3 h), with eight genes, including two AP2-EREBP factors, showing early downregulation followed by significant induction at the later stages of anaerobic development (Figure 5b). It is worth noting that 12 of the 23 TIFY TF-encoding genes in the rice genome were differentially expressed in this study, and three of these are among the most conserved differentially expressed TFs under anoxia (Figure 5b). While AP2-EREBPs, HSF and WRKY TFs have been explored for their role under stress and anaerobic conditions (Hattori *et al.*, 2009; Pucciariello *et al.*, 2012; Narsai and Whelan, 2013), even these do not show the level of conservation observed for genes encoding TIFY factors under anoxia, especially relative to the small size of this family.

The top 32 TFs responsive to anoxia (Figure 5b) were also searched in the Rice Interactions Viewer, and two of these were predicted to have significant interacting proteins (interlogs). The bZIP TF (LOC_Os02 g52780) is predicted to interact with calmodulin 1-1 (LOC_Os03 g20370) and a nuclear-localised Ca-dependent kinase (LOC_Os03 g48270) (Figure 5ci). This is particularly interesting, given that Ca signalling is known to have a significant role in the response to abiotic stress, including anoxia in plants

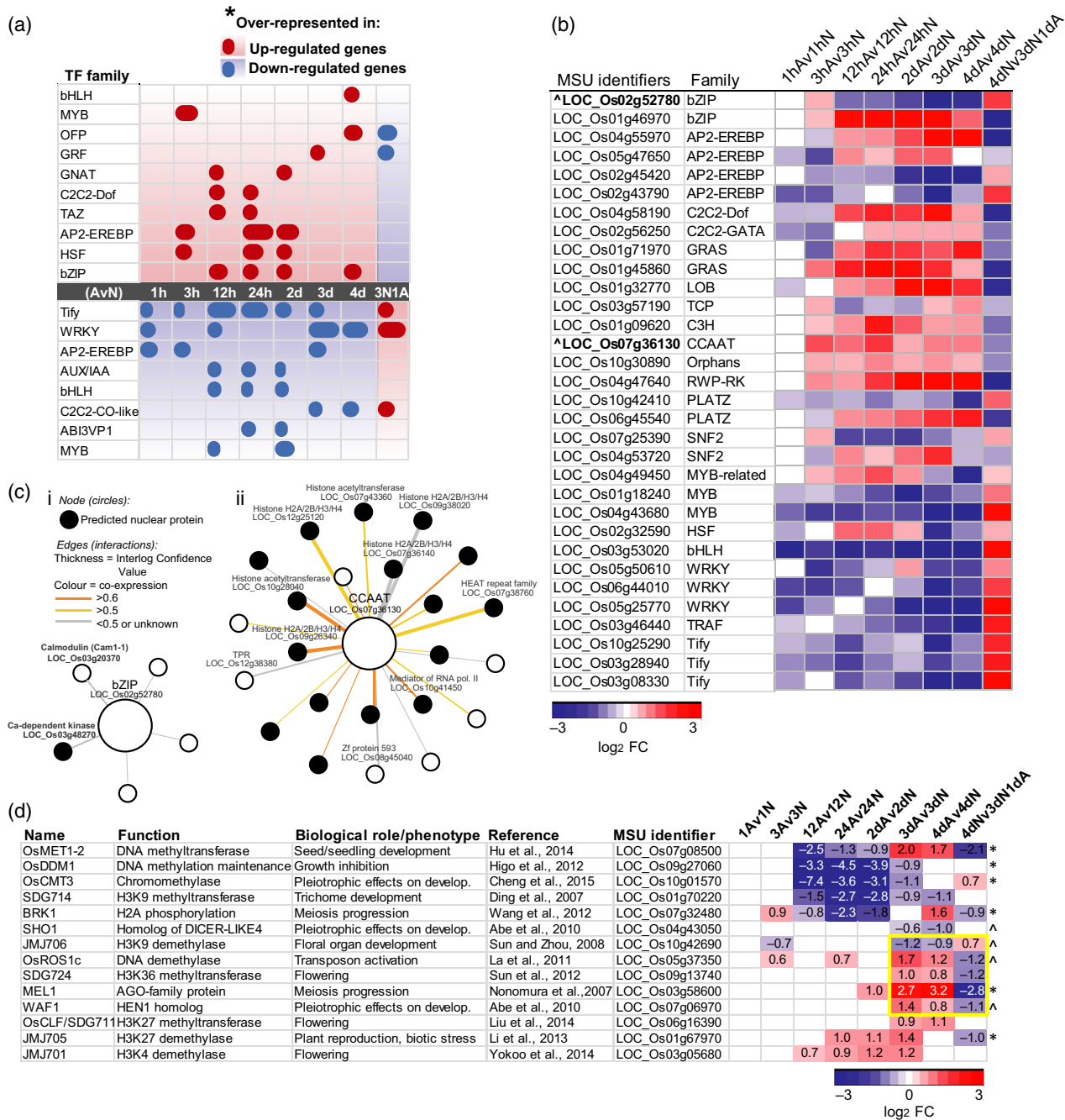


Figure 5. Responses of specific transcription factor (TF) families under anaerobic conditions and after re-oxygenation. (a) Output of the over-representation analysis (Fisher's test, $P < 0.05$) for all TFs for rice in the Plant Transcription Factor Database that were over-represented in at least two time points, comparing aerobically with anaerobically grown (AvN) germinating seed (1, 3, 12 and 24 h) and coleoptile tissue (2, 3 and 4 days), as well as in response to re-oxygenation (4dN versus 3dN1dA). (b) The 32 genes encoding TFs that were differentially expressed in at least seven of the eight comparisons in this study. (c) Two of the 32 genes were found to have predicted interacting proteins (interlogs) using the Rice Interactions Viewer (http://bar.utoronto.ca/interactions/cgi-bin/rice_interactions_viewer.cgi). Only interlogs with medium to high confidence are annotated. The interlog network for (i) the bZIP transcription factor (LOC_Os02 g52780) and (ii) the CCAAT factor (LOC_Os07 g36130) are shown. (d) Differential expression of the functionally characterised genes encoding chromatin modifiers. Genes inside the yellow background were significantly differentially expressed both under anoxic development and upon re-oxygenation. Gene names, functions and biological role are as shown in Shi *et al.* (2014) and Cheng *et al.* (2015). The asterisk (*) indicates that the loss of function of this gene results in reduced fertility/sterility and the hat (^) indicates an altered seed/embryonic phenotype.

(Knight, 2000). A CCAAT factor-encoding gene (LOC_Os07 g36130) was also revealed to have significant predicted interlogs, with most of these encoding core histone proteins or histone acetyltransferases (Figure 5cii). Expression analysis of the histone acetyltransferase genes reveal that five out of six of these genes are induced under anoxia, including the predicted interlog (LOC_Os07 g43360). Given the potential interaction of an anaerobic-responsive CCAAT factor with histone and histone acetyltransferases (Figure 5cii), as well as the known changes that occur in RNA expression (Figure 2) and DNA methylation (Figure 3) under anoxia, we also examined the expression of genes encoding DNA methylation-related functions (Figure 5d). A recent review identified all the functionally characterised genes encoding chromatin modifiers in rice (Shi *et al.*, 2014). These genes, as well as *CMT3* (a more recent addition to this list), were examined for expression changes in this study (Cheng *et al.*, 2015). Of the 26 genes examined, 22 were significantly differentially expressed ($P < 0.05$, $q\text{-value} < 0.05 = \text{FDR} < 5\%$) in response to anoxia, re-oxygenation and/or over germination (Figure S5), with 14 of these differentially expressed at two or more time points under differing oxygen conditions (Figure 5d). Interestingly, it was seen that the loss of function of 10 of these genes resulted in reduced fertility/sterility (indicated by an asterisk) or an altered seed/embryonic phenotype (indicated by ^; Figure 5d). Several of the genes encoding DNA methyltransferases showed significantly lower expression under anoxia (Figure 5d), including *OsMet1* (LOC_Os07g08500), which is responsible for maintenance of CG methylation and was significantly reduced in transcript abundance ($P < 0.05$, $q\text{-value} < 0.05 = \text{FDR} < 5\%$) from 12 h to 2 days under anaerobic conditions, before being upregulated in the 3- and 4-day-old anaerobic coleoptile (Figure 5d). Similarly, *OsCMT3*, which is responsible for CHG methylation, showed significantly reduced expression during germination and in the coleoptile under anoxia, while showing significant induction following re-oxygenation (Figure 5d). While most of the differences in DNA methylation between the aerobically and anaerobically grown coleoptiles were in the CHH context, *OsDRM2* (responsible for CHH methylation) was not differentially expressed in response to differing oxygen conditions but

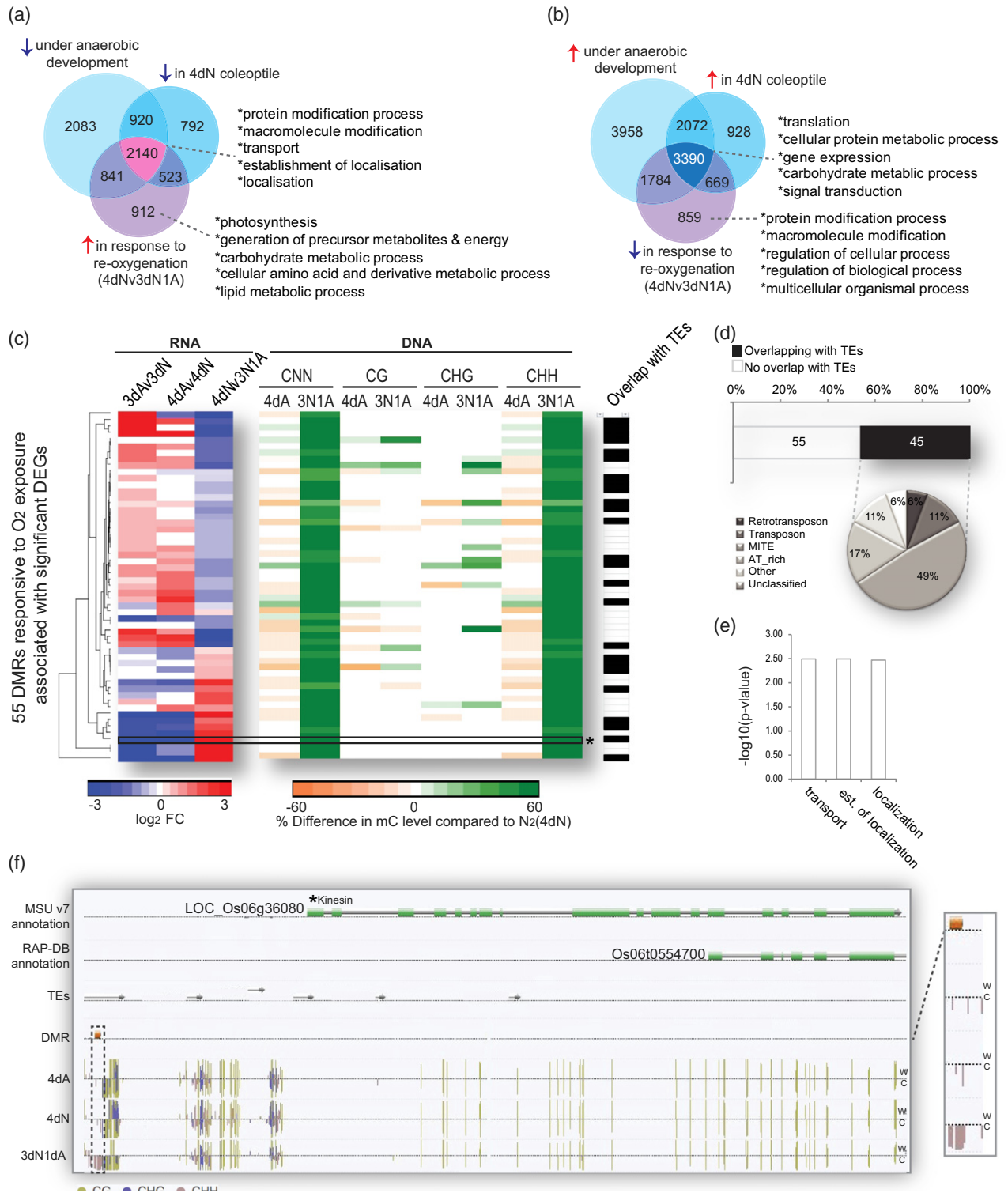
was significantly reduced over the course of germination (Figure S5). Thus, while the changes in DNA methylation under anoxia are largely hyper-methylation in the CHH context, the regulation and maintenance of CG and CHH methylation may also have an important role in seed germination.

Changes in the transcriptome and DNA methylome upon re-oxygenation

After 3 days of growth under anaerobic conditions and 1 day following transfer to aerobic conditions, significant morphological changes were observed (Figure 1). Not surprisingly, significant transcriptomic changes are observed in response to this change (Figure 2a). In contrast to the observations for the changes in DNA methylation (Figure 3), 79% of the upregulated DEGs and 87% of the downregulated genes responsive to re-oxygenation overlapped with the DEGs under anoxia (Figure 6a,b). This analysis revealed 2140 genes that were upregulated under re-oxygenation, downregulated under anaerobic development (up to 3 days) and downregulated in the 4-day anoxic coleoptile (Figure 5a). The top five most significantly enriched categories (GO biological process) in this set were protein and macromolecule modification processes, as well as transport and localisation functions (AgriGO, $P < 0.05$, $\text{FDR} < 0.05$) (Figure 6a). For example, kinases (163 genes), transporters (77 genes) and cellular organisation components (35 genes), such as ankyrin domain-containing proteins and kinesin motor proteins were among these (Figure 6a). In contrast, the 912 genes exclusively induced following re-oxygenation were significantly enriched (AgriGO, $P < 0.05$, $\text{FDR} < 0.05$) in energy generation functions such as photosynthesis, generation of precursor metabolites and energy, carbohydrate metabolic processes and lipid metabolic processes (Figure 6a). Examples of these include genes encoding components of photosystems I and II (eight genes), the mitochondrial ATP synthase (e.g. delta, epsilon, gamma chain; five genes) and subunits of the vacuolar ATP synthase (three genes). This induction of genes encoding energy-related functions is not surprising, given that in the presence of oxygen, greater energy generation is necessary to drive the rapid growth and development that occurs within only 24 h of re-oxygenation.

Figure 6. Molecular responses to re-oxygenation in mature rice coleoptiles.

(a) The transcriptomic response to re-oxygenation [4dNv3dN1dA; 4 days of nitrogen (anaerobic) versus 3 days of nitrogen followed by 1 day of air (anaerobic/aerobic)] was compared with the responses in 4dN coleoptiles (4dAv4dN) and over the entire time course (AvN) with the overlaps shown for (a) upregulated genes and (b) downregulated genes. Gene Ontology (GO) over-representation analysis was carried out for the overlapping common sets of genes, as well as the gene sets exclusively responsive to re-oxygenation. (c) Of the 282 differentially methylated regions (DMRs) identified in the 4dN versus 3dN1dA comparison, 55 were proximal to differentially expressed genes (DEGs). Fold-changes of these 55 genes were hierarchically clustered and shown as a heatmap. Next to this, Cytosine DNA methylation (mC) levels are shown as percentage differences compared with 4dN as a heatmap. DMRs overlapping with transposable elements (TEs) are indicated by black boxes. (d) The percentage of DMRs overlapping with TEs is shown, as well as the breakdown of the types of TEs. (e) Functional categories identified by GO over-representation analysis of the 55 proximal DEGs. (f) An example of a DMR which is proximal to a DEG (4dN versus 3dN1dA) encoding a kinesin protein (LOC_Os06 g36080). A significant difference in CHH methylation ($P < 0.05$; Student's *t*-test) is shown.



Among the downregulated genes following re-oxygenation, 3390 of these genes were also upregulated at one or more time points over anaerobic development (up to 3 days) and in the 4-day anaerobically-grown coleoptile (Figure 6b). These were significantly enriched in several

processes including translation functions (including 73 genes encoding ribosomal proteins and five genes encoding translation initiation factors) as well as gene expression functions, signal transduction and protein and carbohydrate metabolic processes (Figure 6b). In contrast, the 859

genes exclusively downregulated following re-oxygenation were significantly enriched in protein and macromolecule modification as well as biological and cellular regulatory processes, including receptor-like kinases as well as TFs (Figure 6b). Thus, it was revealed that only 13–21% of genes responsive to re-oxygenation are exclusive to this treatment, with the majority of genes showing responses corresponding with those under anaerobic development.

In contrast to the findings from the transcriptomic analysis (Figure 6a,b), distinct changes in DNA methylation were observed following 24 h of re-oxygenation, with 280 out of the 282 DMRs identified showing hyper-methylation compared with the anaerobically grown coleoptile (4dN versus 3dN1dA; Figure S6). Hierarchical clustering of the percentage difference in methylation between re-oxygenated samples and anoxia in each context (CG, CHG and CHH) revealed that the hyper-methylation almost exclusively occurred in the CHH context, with a small subset showing changes in the CHG methylation level as well (Figure S6). It is worth noting that this hyper-methylation was largely specific to re-oxygenation and not simply a reversal of the changes in methylation that occur under anoxia (Figure S6). When the re-oxygenation-responsive DMRs were overlapped with TEs, 53% of the 282 DMRs overlapped (Figure S6), which is 26% lower than that seen for the aerobic-responsive DMRs (Figure S4). Similarly, despite the larger number of re-oxygenation-responsive DMRs (282) compared with anaerobic-responsive DMRs (256), fewer of these (55 DMRs) were proximal to DEGs (Figure 3).

Closer analysis of the 55 DMRs associated with DEGs revealed that 36 of these were downregulated following re-oxygenation (Figure 6c). All 55 re-oxygenation-responsive DMRs showed hyper-methylation compared with the anaerobically grown coleoptile, with most of these also showing hyper-methylation compared with the aerobically grown coleoptile (Figure 6c). As expected, only 45% of the 55 DMRs were overlapping with TEs, and nearly half of these represented MITEs (Figure 6d). When the functions of the 55 DEGs proximal to DMRs were examined, a significant over-representation (AgriGO, $P < 0.05$, FDR < 0.05) of transport and localisation functions was seen (Figure 6e). Specifically, these included 14 cell-associated proteins, several of which are involved in transport and localisation, such as resistance-associated proteins involved in transmembrane movement of substances (e.g. LOC_Os06g08560, LOC_Os12g39180) as well as a trafficking protein (LOC_Os01g33020) and a kinesin motor domain-containing protein (LOC_Os06g36080), among others. The kinesin motor domain-containing protein (LOC_Os06g36080) is induced 3.4-fold in response to re-oxygenation (black box in Figure 6c) and visualising this reveals a DMR in the upstream region of this gene, with significant hyper-methylation (FDR < 0.01) observed upon re-oxygenation, particularly in the CHH context (Figure 6f). Unlike the

transcriptome, where there is large overlap with the anaerobic response (Figure 6a,b), there appears to be a re-oxygenation-specific change in DNA methylation that is largely distinct from the aerobically and anaerobically grown coleoptiles (Figure S4). Taken together with the fact that the DEGs proximal to DMRs included both up- and downregulated genes, there is no evidence for a simple correlative relationship between DNA methylation and gene expression.

Conserved patterns in the transcriptomes and methylomes during germination

Given that germination and early development progresses successfully under both aerobic and anaerobic conditions in rice, we hypothesised that at least some transcripts are likely to respond in a similar manner over the entire time course. In order to test this, the RNA-seq data were compared between dry seeds, the 24-h germinated embryos and the 4-day coleoptiles under aerobic and anaerobic conditions separately (Figure 7ai, aii). Of all the DEGs between dry seeds and the 24-h anoxia-germinated embryo, 82% of genes overlapped with the DEGs in the aerobic-germinated embryo compared with dry seed (overlapping genes; Figure 7ai). Furthermore, step-wise comparisons were carried out over the time course and all significant DEGs ($P < 0.05$, q -value $< 0.05 = \text{FDR} < 5\%$) were analysed for over-represented functional categories (Usadel *et al.*, 2006). This analysis revealed that, independent of oxygen conditions, cell wall, hormone metabolism, stress and signalling functions are among the common upregulated genes as these were over-represented in at least four comparisons (Figure 7b). The GRAS family of TFs was over-represented among the up-regulated genes over germination, from the early time-points to 12–24 h post-imbibition, independent of oxygen conditions (Figure 7b). Interestingly, whilst signalling was over-represented in the upregulated genes at the later stages of both aerobic and anaerobic germination; the step-wise comparison reveals that these were also induced during early anaerobic germination (i.e. 0 h versus 1 h and 1 h versus 3 h), whilst this was not seen under aerobic germination (Figure 7b). A similar pattern is also seen for the downregulated genes under both aerobic and anaerobic germination, with an over-representation of lactoylglutathione lyse-encoding genes involved in the biodegradation of xenobiotics, as well as late embryogenesis abundant (LEA) proteins observed from the earliest comparisons under anaerobic germination (i.e. 0 h versus 1 h and 1 h versus 3 h; Figure 7b). In contrast, calcium signalling was over-represented among the upregulated genes at the later stages of both aerobic and anaerobic germination; notably, under aerobic germination this was also seen at the earlier stages (i.e. 0 h versus 1 h and 1 h versus 3 h) unlike anaerobic germination (Figure 7b). Thus, these findings support the proposed hypothesis revealing

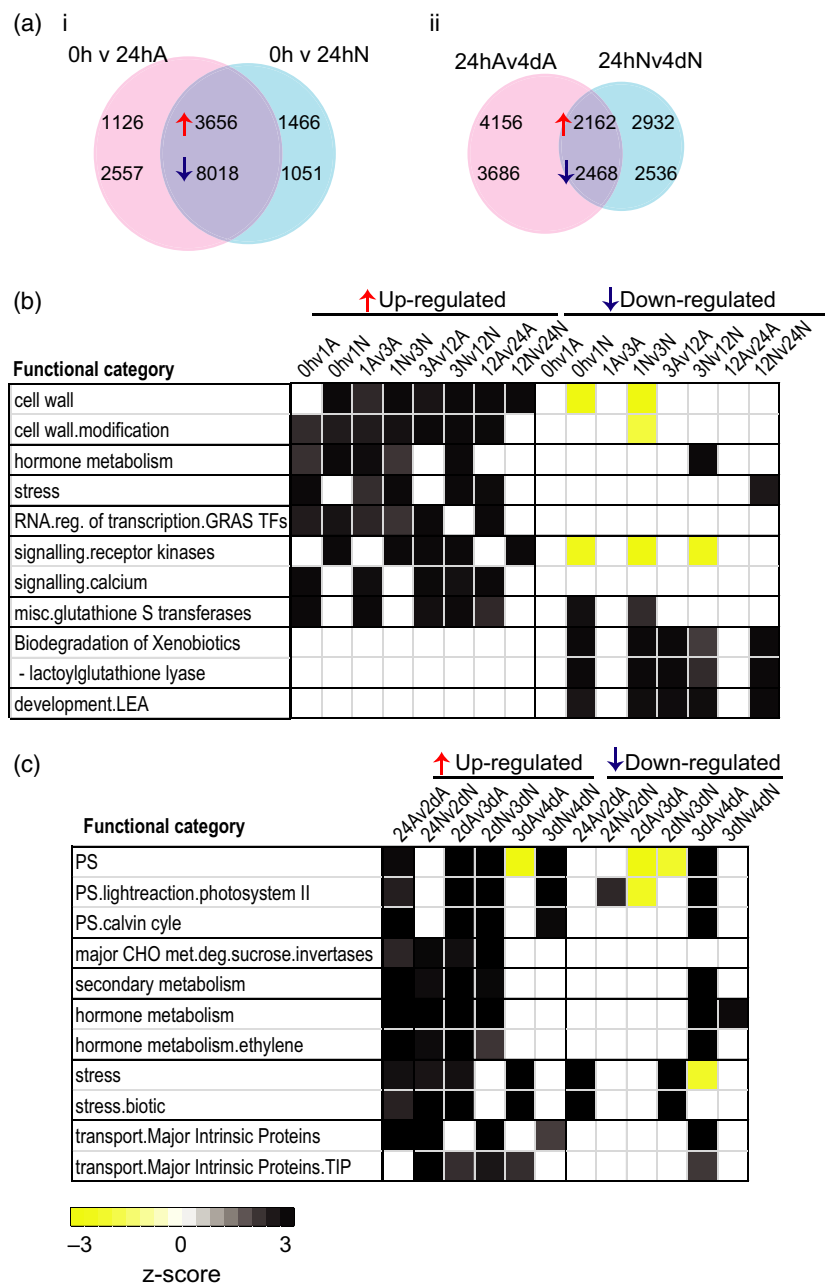
some conservation in the transcriptomic responses over the course of germination, independent of the presence of oxygen, as well as revealing several stage-specific differences over aerobic and anaerobic development.

Unlike the germination comparisons, only 46% of all the DEGs between the 24-h seedling and 4-day coleoptile under anoxia overlapped with the same comparison under aerobic conditions (Figure 7aii). This is not entirely unexpected, as there are also more obvious morphological differences in these tissues at 4 days, such as radicle emergence under aerobic conditions, which is not seen in the 4-day anaerobically grown seedling (Figure 1b). Nevertheless, analysis of

the over-represented functional categories in the step-wise comparisons from germinated seed to mature coleoptile did revealed several similarities (Figure 7c). Genes encoding proteins involved in photosynthesis were upregulated in the comparisons from 24 h to 3 days under aerobic conditions, whereas this is slightly delayed in under anaerobic conditions, largely occurring from 2 to 4 days (Figure 7c). In contrast, the over-representation of major carbohydrate metabolism functions, specifically sucrose invertase, as well as secondary metabolism and hormone metabolism (ethylene) was conserved in the upregulated gene-sets from 24-h- to 3-day-old coleoptiles under both aerobic and anaerobic

Figure 7. Transcriptomic changes during germination and coleoptile development.

(a) (i) Venn diagram comparing differentially expressed gene sets for the dry seed embryo (0 h) versus 24-h comparisons, revealing the number of genes showing common responses independent of oxygen availability. (ii) Venn diagram comparing differentially expressed gene sets for the germinated seed 24 h versus 4 day comparisons. A, aerobic/air conditions; N, anaerobic/nitrogen conditions. (b) Over-represented functional categories in the step-wise comparisons over the germination time course (0 to 24 h). Significance is indicated by a z-score, where the highest significances are coloured black. (c) Over-represented functional categories in the step-wise comparisons from germination to mature coleoptile (24 h to 4 days).



conditions (Figure 7c). While stress and transport functions were over-represented in the upregulated genes sets under both aerobic and anaerobic conditions, certain time-specific differences were seen, such as the over-representation of transport functions in all comparisons from 24-h- to 4-day-old anaerobic seedlings, while this was only seen between 24 h and 2 days in the aerobically grown seedlings (Figure 7c). It is worth noting that photosynthesis, secondary metabolism and hormone metabolism functions were over-represented in the downregulated gene-set from 3- to 4-day-old aerobically grown coleoptiles, indicating that despite earlier upregulation, this is not maintained in mature aerobically grown coleoptiles and is different from that seen in anaerobically grown coleoptiles (Figure 7c).

Like the transcriptome (Figure 7ai), the DNA methylome was also highly conserved during germination, with no significant DMRs (FDR < 0.01) identified between the 24-h aerobically and anaerobically germinated seed (Figure 3). Given that DNA methylation was also analysed in the dry

seed, it was possible to compare the dry seed methylome with the 24-h germinated seeds to reveal any changes in DNA methylation over rice germination. In this way, 1321 DMRs were identified over the course of germination (independent of oxygen) and all of these were significantly hypo-methylated (FDR < 0.01) at 24 h compared with the dry seeds at 0 h (Figure S7). Of the 700 DMRs, 53% overlapped with known repeat and TE regions (Figure S7), and 435 were proximal to DEGs (Figure 8a). When the 435 DEGs associated with the germination-induced DMRs were analysed further, over 85% showed similarities in the fold-change responses at 24 h under aerobic and anaerobic conditions compared with dry seeds, and all of these were associated with significant hypo-methylation (Figure 8b). Over-representation analysis also revealed that these 435 genes were significantly enriched (AgriGO, $P < 0.05$, FDR < 0.05) in transport, localisation and regulation functions, with nearly 20% of these encoding proteins with DNA/RNA-binding functions (Figure 8c).

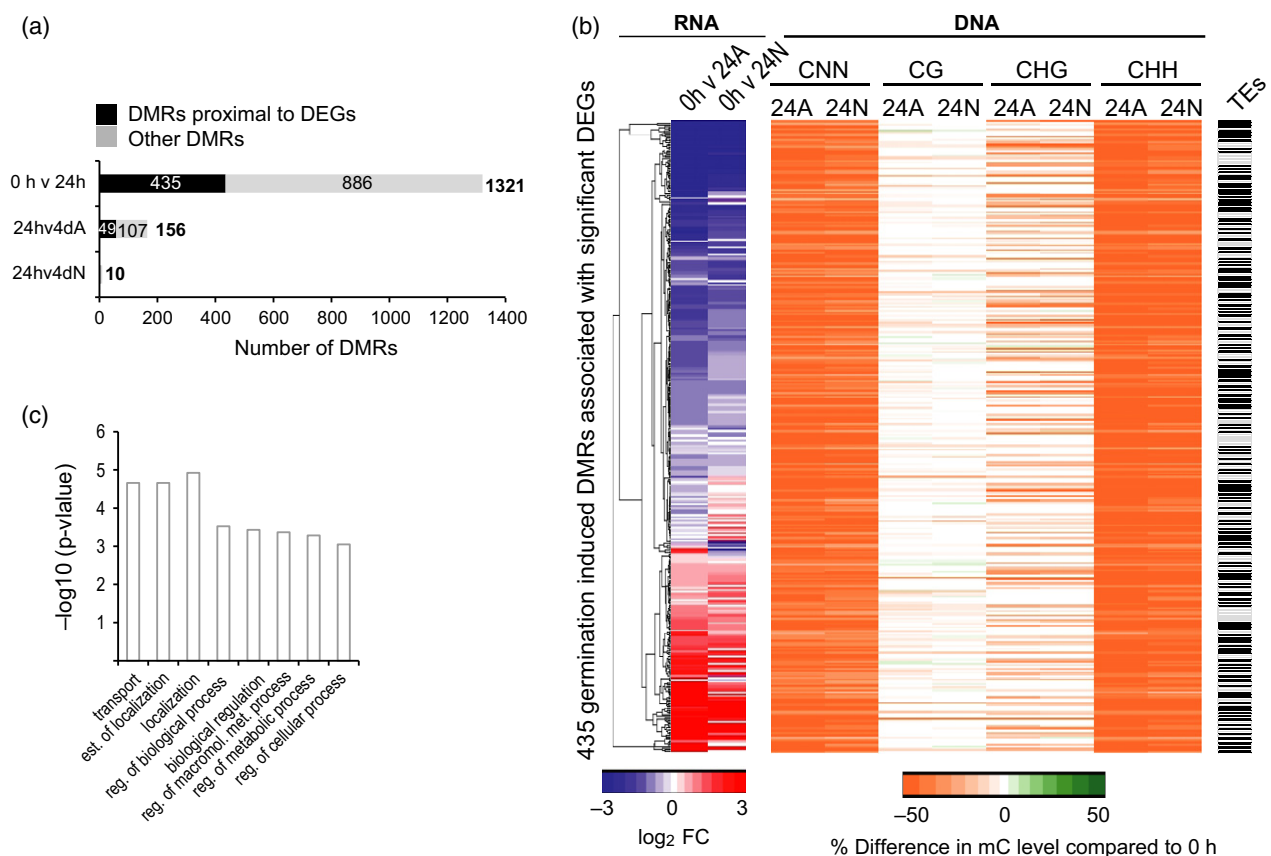


Figure 8. Differential methylation during germination.

(a) Summary of the number of differentially methylated regions (DMRs) during germination and coleoptile development. For each of the DMRs, proximal genes were also identified and the number DMRs proximal to differentially expressed genes (DEGs) are indicated in black.

(b) Fold-changes of the 435 DEGs that were proximal to DMRs were hierarchically clustered and are shown as a heatmap. Next to this, mC levels are shown as percentage differences compared with 4 days of aerobic conditions (4dA) as a heatmap. DMRs overlapping with transposable elements (TEs) are indicated by black boxes.

(c) For the 435 DMRs identified in the dry seed (0 h) versus 24 h comparison, the functional categories identified by Gene Ontology (GO) over-representation analysis are shown.

Unlike germination, the comparison of 24-h germinated seeds with mature coleoptiles revealed a divergence in both the transcriptomic (Figure 7a) and methylome comparisons under aerobic and anaerobic conditions (Figure 8a). In fact, only 10 DMRs were identified between the 24-h seedling and 4-day-old coleoptile under anoxia, and none of these overlapped with the 156 DMRs identified in the equivalent comparison under aerobic conditions (Figure 8a). Of these 156 DMRs, 151 were hypo-methylated at 4 days compared with 24 h under aerobic conditions, continuing the trend of de-methylation from dry seeds to 24-h A (Figure 8a).

Re-oxygenation results in methylation changes similar to the dry seed methylation state

Whilst examining the DMRs induced during germination, it was apparent that several of these were identified in response to re-oxygenation. To compare both sets, the DMRs identified in the 0 h versus 24 h and the 4dN versus 3dN1dA comparisons were compared (Figure 9a). Two-thirds (188) of all the DMRs identified in response to re-oxygenation overlapped with the germination-responsive DMRs (Figure 9a). As previously noted, significant differential methylation was observed upstream of a kinesin-encoding gene (LOC_Os06 g36080) in response to re-oxygenation (Figure 6f). This DMR was also hyper-methylated in dry seeds, and this is lost upon imbibition in the 24-h germinated seed as well as the 4-day-old coleoptile under both aerobic and anaerobic conditions (Figure 9b). However, after 1 day of re-oxygenation this region is again hyper-methylated, similar to the levels seen in dry seeds (Figure 9b). It is worth noting that similar to the DMR upstream of the kinesin gene (Figure 9b), all of the 188 DMRs identified over germination and in response to re-oxygenation showed similar hyper-methylation in dry seeds and again upon re-oxygenation (Figure 8a).

Based on the differential methylation analyses carried out in this study, it was revealed that, independent of oxygen conditions, changes in DNA methylation are conserved between dry seeds and 24 h after imbibition, revealing over 1000 hypo-methylated regions upon imbibition (Figure S7). Following 24 h, a further 151 out of the 156 DMRs were hypo-methylated under aerobic conditions, whilst only 4 of the 10 DMRs were hypo-methylated under anaerobic conditions in the 24-h versus 4-day coleoptile comparisons. When the 4-day coleoptiles were compared, 256 DMRs showing hyper-methylation under anaerobic conditions were observed compared with the aerobic coleoptile. Thus, when re-oxygenation occurs after 3 days under anaerobic conditions, it might be expected that methylation levels change in a manner similar to that seen in the aerobic coleoptile, as this is the trend followed by the transcriptome. However, the opposite is seen, whereby hyper-methylation of 282 regions was revealed, similar to

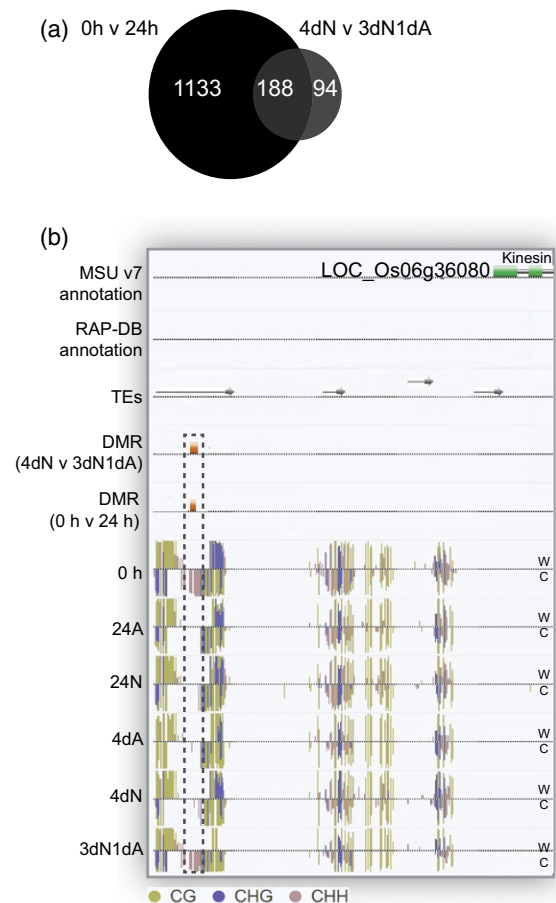


Figure 9. Re-oxygenation results in methylation patterns more similar to those of dry seed.

(a) Venn diagram showing the overlapping differentially methylated regions (DMRs) proximal to differentially expressed genes (DEGs) identified between the 0 h versus 24 h comparison and the 4-day anaerobic (4dN) versus 3 days of anaerobic followed by 1 day of aerobic conditions (3dN1dA) comparison.

(b) An example of an overlapping DMR (0 h versus 24 h AND 4dN versus 3dN1dA), which is proximal to a DEG encoding a kinesin protein (LOC_Os06 g36080). Significant increases in CHH methylation ($P < 0.05$; Student's *t*-test) are seen at 0 h and upon re-oxygenation at 3dN1dA.

that observed in dry seeds. This unexpected overlap between the DMRs identified in response to re-oxygenation and germination shows cross-tissue conservation and supports a functional role for the epigenetic regulation of these regions. In plants, the association between methylation levels and specific genomic features is generally conserved, with TEs often showing high levels of methylation (in all contexts) in rice, which already has a higher TE content (>35%) in the genome (Secco *et al.*, 2013; Xing *et al.*, 2015) compared with Arabidopsis. Here, the high overlap (>75%) with TEs was only observed in the anoxia-induced DMRs identified between the 4-day coleoptiles (Figure 4), while <50% of the re-oxygenation-responsive DMRs and germination-induced DMRs overlapped with TEs (Figure 6

& 8), indicating that these DMRs require further exploration as they may be acting via a different regulatory mechanism.

DISCUSSION

In rice, germination and seedling development can progress in both the presence and absence of oxygen. While the transcriptomic responses to anaerobic conditions, found using microarrays, have revealed common responses, even across kingdoms, these alone are not sufficient for altering tolerance to anaerobic conditions (Lasanthi-Kudahettige *et al.*, 2007; Narsai *et al.*, 2009; Mustroph *et al.*, 2010; Bailey-Serres *et al.*, 2012). Therefore we undertook an in-depth study, at several time points during germination and in the rice coleoptile using RNA-seq analyses. As all the rice factors known to date that give greater anaerobic tolerance – SUB1A (Xu *et al.*, 2006), SNORKEL (Hattori *et al.*, 2009) and trehalose-6-phosphate phosphatase (Kretzschmar *et al.*, 2015) – are regulators of expression or metabolism, we hypothesised that an in-depth transcriptome would reveal features missed in previous studies and offer insights into the molecular responses in rice that allow it to germinate under anaerobic conditions. Additionally, there are significant questions about the potential role of DNA methylation during germination, in response to anoxia, and how these may link with transcriptomic responses, particularly in the context of time- and tissue-specific regulation. Thus, the DNA methylome was monitored in parallel to determine if it changed under these conditions, especially considering that changes in the epigenome are generally slow and maintained whereas the response to oxygen limitation and re-oxygenation are rapid.

Several mutant studies have shown an essential role for epigenetic regulation in the progression of normal seed germination and early development (Moritoh *et al.*, 2012; Hu *et al.*, 2014; Shi *et al.*, 2014; Yamauchi *et al.*, 2014; Cheng *et al.*, 2015). In the current study, genome-wide epigenetic analysis revealed the significant and almost identical hypo-methylation occurs over both aerobic and anoxic germination, with over 1000 DMRs identified (FDR < 0.01), mostly in the CHH context between dry seeds and germinated seedlings (0 h versus 24 h; Figure 8a). For the transcriptome, more the double the number of DEGs were identified here compared with previous microarray studies during germination (Howell *et al.*, 2009; Narsai *et al.*, 2009) and in coleoptiles (Lasanthi-Kudahettige *et al.*, 2007; Narsai *et al.*, 2015), with over 1000 putative unannotated DEGs also identified here (Figure 2). Interestingly, over 80% of DEGs overlapped between aerobic and anoxic germination (0 h versus 24hA and 0 h versus 24hN; Figure 7ai), revealing conservation in both the overall transcriptomic response and the epigenetic responses over the course of seed germination. These overlapping DEGs included the

significant ($P < 0.05$, $q\text{-value} < 0.05 = \text{FDR} < 5\%$) induction of *OsMET1-2* (responsible for maintaining CG methylation) and reduction of *OsDRM2* (responsible for CHH methylation) over both anoxic and aerobic germination (Figure S5), and corresponding with the pattern of changes in DNA methylation during germination. Notably, the loss of function of both of these genes results in abnormal development/seedling lethality in rice (Moritoh *et al.*, 2012; Hu *et al.*, 2014). Thus, it is proposed that epigenetic and transcriptomic regulation during seed germination in rice (0–24 h) largely occurs in an oxygen-independent manner.

While there are clearly differences in the abundance of some transcripts, the underlying developmental process appears to be conserved, and the fact that there is no significant difference in the 24hA and 24hN methylomes is a reflection of this similarity. Also, importantly, the lack of oxygen does not trigger a stress response in rice, and if anything the stress response seems to be suppressed. Our transcriptomic analysis revealed an early and consistent suppression of the stress-responsive WRKY and TIFY TFs and induction of bZIP TFs under anoxia in rice (Figure 5a). Previous analyses have revealed the opposite, i.e. induction of WRKY factors in Arabidopsis under hypoxia (Narsai and Whelan, 2013), and sequence analyses have also revealed that rice TIFY TFs contain motifs beyond the characteristic Arabidopsis TIFY domain (Ye *et al.*, 2009). Furthermore, while the N-end rule pathway has been shown to be crucial in the hypoxia response of Arabidopsis, the *SUB1A* gene (encoding a group VII ERF TF), which is a major factor in tolerance of submergence in rice, has been shown not to be a substrate of this pathway (Gibbs *et al.*, 2011). Thus, this indicates some divergence in the mechanisms of low oxygen tolerance between Arabidopsis and rice. For example, the recently identified HRU1 is induced under low oxygen and has been shown to be a key regulator of a ROS signalling network required for tolerance of low oxygen in Arabidopsis (Gonzali *et al.*, 2015), whereas the rice orthologue of this gene does not appear to be induced under hypoxia (examined using RiceDB; Narsai *et al.*, 2013). Thus, in addition to understanding how rice withstands anaerobic conditions, it may be as important to characterise what it does not do, i.e. turn on a stress response that could be detrimental at this crucial stage of development and shift valuable resources from growth to stress responses. Furthermore the conservation of the DNA methylome at 24 h (A and N) may explain the greater sensitivity and sterility/seedling lethal phenotypes of DNA methyltransferase mutants in rice (Moritoh *et al.*, 2012; Hu *et al.*, 2014; Yamauchi *et al.*, 2014). Also, like the suppression of the transcriptome stress response, it may be crucial that the germination methylome is maintained in a defined state, which if altered is not compatible with the unique developmental profile that occurs under both aerobic and anaerobic conditions. Additionally, the finding that re-

oxygenation of 3-day-old coleoptiles results largely in a reversion of the methylome to that observed in dry seeds indicates cross-tissue conservation, supporting a functional role for the epigenetic regulation of these regions and indicating that this basic pattern is important for development.

Following germination, we revealed over 250 significant DMRs ($FDR < 0.01$) and thousands of DEGs ($P < 0.05$, q -value $< 0.05 = FDR < 5\%$) between the 4-day aerobic and anoxic coleoptiles (Figures 2 and 4), indicating that oxygen-dependent epigenetic and transcriptomic changes occur following germination in rice seedlings. In fact, the extensive transcriptomic analyses revealed an increasing number of DEGs as the time course progressed, with the greatest transcriptomic differences seen in the mature coleoptiles. Interestingly, only 10 DMRs were identified between the 24-h aerobically/anaerobically grown embryos and the 4-day anaerobically grown coleoptiles, thus the epigenetic differences in the 4-day aerobic and anaerobic coleoptiles largely reflect the de-methylation specifically occurring in the aerobically grown coleoptile (Figure 4). Despite the significant transcriptomic differences observed from 24 h embryos to 4-day old coleoptiles under anoxia (Figure 7a), the observed conservation of the epigenome may reflect differences in development that become obvious after germination, whereby cellular maintenance and elongation are largely observed in the anoxic coleoptile, while the cells of the aerobic coleoptile have a greater rate of cell division and begin senescing after 4 days as development progresses. Under aerobic conditions, greater energy supply enables more cell division and cell specialisation to occur as primary root development and radicle protrusion occur after 2 days, while cell elongation in the coleoptile accelerates more rapidly under anoxia (Figure 1) (Takahashi *et al.*, 2011; Edwards *et al.*, 2012; Narsai *et al.*, 2015). Under both aerobic and anaerobic conditions, cell division begins at 24 h to 2 days after imbibition, though to a lesser extent under anoxia, before gradually ceasing by 3 days as cell elongation occurs in the mature rice coleoptiles (Wada, 1961; Öpik, 1973; Takahashi *et al.*, 2011; Edwards *et al.*, 2012; Narsai *et al.*, 2015). This also corresponds with the transcriptomic analyses which reveals a common, oxygen-independent upregulation of cell wall functions during germination (unit 24 h; Figure 7b), which was not maintained after 24 h as seedling morphology begins to diverge under the differing oxygen conditions (Figure 1).

The expression of *OsMET1-2* and *OsCMT3*, which are responsible for maintenance of CG and CHG methylation, respectively, during DNA replication, also remained significantly lower during germination and in the 2-day coleoptiles under anoxia (Figure 5d), corresponding with the aforementioned timing and reduced cell division known to occur under anoxia. Mutants for *OsMET1-2* (Hu *et al.*, 2014) and *OsCMT3* (Cheng *et al.*, 2015) have been shown

to have significant developmental and morphological abnormalities, confirming an essential role for epigenetic regulation in normal morphological development in rice. A recent study has also shown significant differences in the transcriptome along the length of the developing rice coleoptile, with differential expression observed between the bases and tips of both aerobically and anaerobically grown coleoptiles (Narsai *et al.*, 2015). Interestingly, cell division and elongation has also been shown to be significantly reduced in the alcohol dehydrogenase mutant, resulting in reduced coleoptile elongation under submergence (Takahashi *et al.*, 2011) and confirming the link between the regulation of cellular development and anoxic morphology in young rice seedlings. After only 24 h of re-oxygenation, significant changes in seedling morphology are also observed, including primary root and leaf development (Figure 1), similar to that seen under aerobic conditions; not surprisingly, this is accompanied by significant restoration of transcriptomic responses towards aerobic development (Figure 5a,b). Previous studies in rice have observed post-stress recovery following phosphate starvation (Secco *et al.*, 2015) and salt stress (Wang *et al.*, 2015); however, despite transcriptomic reversions, most of the epigenetic changes remained even after up to 7 days of recovery in both studies. Here, we show that significant changes in both DNA methylation and seedling morphology are seen within only 24 h of re-oxygenation, revealing very rapid epigenomic regulation at a rate that has, to our knowledge, not previously been seen in rice. The significant overlap of these DMRs with the dry seed epigenome suggests that this may be involved in regulating rapid changes in rice growth.

EXPERIMENTAL PROCEDURES

Plant growth conditions

Rice (*Oryza sativa* 'Amaroo') was de-hulled, surface sterilised and grown in 250 ml of culture medium [0.5 mM 2-(*N*-morpholine)-ethanesulphonic acid (MES), 0.4 mM CaSO_4 , pH 6.5] that was bubbled with air or N_2 (6–7 L min^{-1}) gas, as described previously (Shingaki-Wells *et al.*, 2011). About 50–75 seeds were placed into each medium-containing flask and these were grown for 1 h, 3 h, 12 h, 24 h, 2 days, 3 days and 4 days under aerobic (A) and anaerobic (N) conditions at 30°C in the dark, before being harvested separately at each time point. In addition, 3-day-old seedlings grown under anaerobic conditions were switched to air for 1 day in order to generate the re-oxygenation samples (3dN1dA). For the dry seed and time points up to 24 h, the embryos were rapidly dissected and snap frozen, as done previously (Narsai *et al.*, 2009), and for all time points after that (2, 3 and 4 days and 3dN1dA) only whole coleoptiles were sampled and snap frozen. For the later time points (under aerobic conditions) primary leaf tissue was removed and only coleoptiles were harvested.

RNA isolation and RNA sequencing methods

Total RNA for all samples was isolated using the Sigma Spectrum Plant Total RNA kit (<http://www.sigmaaldrich.com/>). RNA integrity

and quality were checked using the NanoDrop 1000 Spectrophotometer (<http://www.nanodrop.com/>) and gel electrophoresis with an Agilent Bioanalyser (<http://www.agilent.com/>). Only RNA samples with Abs260 nm/Abs 280 nm ratios of 2–2.1 were used for further analysis. Three biological replicates were used per sample (representing each time point) for RNA-seq library generation, following Ribo-Zero rRNA removal (Epicentre, <http://www.illumina.com/>) as described previously (Secco *et al.*, 2013). Illumina TruSeq Stranded Total RNA with Ribo-Zero Plant Sample Prep Kits were used to generate the libraries and six RNA-seq libraries were multiplexed and loaded per lane in the Illumina HiSeq flow cell v3. All sequencing procedures were carried out as per the manufacturer's instructions (Illumina, <http://www.illumina.com/>) using the Illumina HiSeq 1500. Quality control was confirmed within the Illumina HiSeq software and all reads passing filter were mapped onto the International Rice Genome Sequencing Project 1.0 (IRGSP-1.0) rice genome sequence (from the RAP-DB).

Genomic DNA extraction and MethylC-seq library generation

The Qiagen DNeasy Plant Minikit was used to extract genomic DNA and 600 ng of purified genomic DNA was used for MethylC-seq library preparation after spiking in 0.5% λ DNA (N_6 -methyladenine-free) (New England BioLabs, <https://www.neb.com/>). MethylC-seq libraries were prepared as described previously (Lister *et al.*, 2013; Secco *et al.*, 2015) and three biological replicates were used per sample (each time point). First, genomic DNA was sonicated (Covaris, <http://covarisinc.com/>) to an average of 200 bp, this was end-repaired (End-It DNA End Repair Kit, Epicentre) and single-read methylated adapters were ligated (Illumina). The MethylCode Bisulfite conversion kit (ThermoFisher Scientific, <http://www.thermoFisher.com/>) was used for sodium bisulphite conversion, as per the manufacturer's instructions. Bisulfite-converted, adapter-ligated DNA molecules (20 μ l) were mixed with 25 μ l of KAPA HiFi Hotstart Uracil+ ReadyMix (Kapa Biosystems, <https://www.kapabiosystems.com/>) and 5 μ l of TruSeq Primer Cocktail (Illumina) before PCR amplification (initial denaturing at 95°C for 2 min, then four cycles of 98°C for 20 sec, 60°C for 15 sec and 72°C for 1 min, followed by 10 min at 72°C) and purification using AMPure XP beads (Beckman Coulter, Brea, CA, USA). This was sequenced for 51 cycles using the Illumina HiSeq 1500, following the manufacturer's instructions.

RNA-seq data analysis

The cufflinks package (Trapnell *et al.*, 2012) version 2.2.0 (Bowtie2 v.2.2.0-beta7 and Tophat v.2.2.0) was used for all RNA-seq analyses, carried out as previously (Lister *et al.*, 2013; Secco *et al.*, 2013). Basically, the rice IRGSP-1.0 genome sequence (from the Rice Annotation Project database, <http://www.rap.com.jp>) was used to generate a Bowtie2 index and TopHat was used for aligning the RNA-seq reads to the genome using the gene model annotation file (GFF) from the MSU Rice Genome Annotation Project database v.7 (<http://rice.plantbiology.msu.edu/>) with the following options: `-b2-sensitive -r 0 -mate-std-dev 100 -g 1 -G`. The RABT assembly was generated using the resulting aligned reads with Cufflinks v.2.2.0 in discovery mode, in order to identify previously unannotated genes (as previously described; Trapnell *et al.*, 2012). To do this, reads corresponding to the dry seed, 24hA, 2dN, 4dA and 4dN were merged using Cuffmerge and the MSU v.7 gene model annotation file as the reference annotation to create this new RABT assembly for transcript abundance quantification. Transcript abundance (fragments per kilobase of transcript per million mapped reads, FPKM) and differential expression analysis were determined using Cufflinks under default parameters ($P < 0.05$

and $q < 0.05 = 5\%$ FDR cut-off) as well as incorporating upper quartile normalisation and bias detection and correction ($-N -b$). The fold-changes produced represent $\log_2$ of the expression ratio, e.g. (FPKM(4dN)/FPKM(4dA)) for the 4dA versus 4dN comparison. Comparison of the expression responses for 24-h germinated seedlings and 4-day-old coleoptiles from previous studies confirm correlation (up to 64% overlap) between the expression responses in this study with previous studies (Figure S8).

The AnnoJ genome browser was used to visualise the mapped RNA-seq reads (<http://www.anno.j.org/>) and the data from this study can be found at: http://www.plantenergy.uwa.edu.au/anno.j/Reena_Rice_AN_RNA_mC.html. The annotations loaded include tracks for the MSUv7, RAB-DP and the generated RABT assemblies. By default, all annotation and sample tracks will be loaded, so to avoid the longer time taken to display all of these users can select which tracks to display.

For Figure 5(b), the top 32 anoxia-responsive TFs were searched in the Rice Interactions Viewer (http://bar.utoronto.ca/interactions/cgi-bin/rice_interactions_viewer.cgi), which houses a database of predicted and confirmed rice interacting proteins.

Over-representation analysis was carried out using AgriGO, which relies on Fisher's test for over-representation; significance was set at the default of $P < 0.05$ with a FDR $< 5\%$ (Benjamini-Hochberg < 0.05). More detailed over-representation analyses were carried out using the PAGEMAN tool (Usadel *et al.*, 2006), which also relies on Fisher's test, where significantly over-represented categories had $P < 0.05$.

MethylC-seq data analysis

Using the rice IRGSP-1.0 reference genome, reads were mapped and analysed as previously for MethylC-seq data (Lister *et al.*, 2011, 2013) and DMRs were identified in the same manner as previously described for this data type (Secco *et al.*, 2015). The frequency of cytosine basecalls at reference cytosine positions in the spiked-in lambda genome was normalized against the total number of base-calls at reference positions in the lambda genome (which is unmethylated), enabling the calculation of the bisulphite non-conversion frequency. If regions of differential methylation were within 200 bases of each other and showed the same direction of change in methylation (hypo-/hyper-methylation), these were combined into a block and defined as one DMR. Only DMRs showing significant differences in all three replicates of each sample being compared were considered for further analysis. DMRs were identified as significant where the FDR < 0.01 , with the P -value cutoff set to the maximum that still limited a FDR of < 0.01 , similar to previous studies (Secco *et al.*, 2015). For the analyses presented here, only blocks containing five or more differentially methylated sites were considered significant.

ACCESSION NUMBERS

For all samples, the Illumina reads have been submitted to the Sequence Read Archive at the National Center for Biotechnology Information (<http://www.ncbi.nlm.nih.gov/sra>) under the experiment titled 'RNAseq and DNA methylomes of germinating rice and developing coleoptiles'.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. Examples of different loci identified.

Figure S2. Analysis of putative un-annotated genes.

Figure S3. Analysis of the putative unannotated locus XLOC_014344.

Figure S4. Differential methylation between the aerobically and anaerobically grown coleoptiles.

Figure S5. Differential expression of the functionally characterised genes encoding chromatin modifiers in rice.

Figure S6. Differential methylation between the re-oxygenated coleoptiles and anaerobically grown coleoptiles.

Figure S7. Differential methylation between 0-h (dry seeds) 24-h samples (in air and nitrogen).

Figure S8. Comparison of the differentially expressed genes in this study (using cv. Amaroo) and previous studies in rice.

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